

AI & GenAI Engineering Experience

A curated view of Bill's hands-on AI work — from deploying and evaluating LLM API endpoints to building GenAI workflows in production and advising on ML/MLOps practice — plus his personal practice building with Claude as an agentic development partner. Ordered most recent first.

PROFESSIONAL ENGAGEMENTS

Deloitte consulting engagements with a direct AI/ML/GenAI component. Client names withheld per standard consulting confidentiality practice, consistent with the main CV.

Australian State Government Insurance Company

MAR 2025 – JUN 2025

Data Analyst / Engineer — GenAI-Powered Legacy Migration

A large-scale migration of on-premise data into a new Snowflake Enterprise Data Warehouse required converting a huge volume of legacy SAS scripts into SQL — a slow, error-prone task by hand.

- Built a GenAI workflow using Anthropic's Claude 3.7 to convert legacy SAS scripts into Snowflake SQL at scale
- Owned PII-masking across hundreds of SAS files, designing a Python Regex-based solution to identify and mask/map sensitive data
- Supported the ETL team writing SQL stored procedures across 3 staging layers and 1 target layer to migrate data into production

Claude 3.7

Python (Regex)

Snowflake SQL

SAS

Impact — Delivered productionised code in 50% less time than planned and a 50% reduction in resourcing cost — efficient enough that Bill was moved onto the ETL stream to support further activities off the back of it.

Big 4 Australian Bank

AUG 2024 – SEP 2024

Business Analyst — Generative AI Risk & Compliance MVP

16,000+ compliance incidents are reported annually at this bank; 75% require a compliance assessment taking 2.5–5 hours to complete by hand. The client needed a GenAI copilot to cut that time down.

- Gathered requirements and translated them into delivery, bridging gaps between data scientists, solution architects, and Risk & Compliance business stakeholders
- Ran end-to-end testing and drove sign-off on production-gating artefacts (Test Plan, Test Approach, Detailed Technical Design, Test Completion Report)
- Ran daily standups and scrum ceremonies — backlog grooming, RAID tracking — to keep the build on schedule

GenAI Copilot

Confluence / Jira

Agile / Scrum

Impact — Shipped the first of 5 pioneer GenAI use cases piloted into production at the bank, establishing a repeatable GenAI solution pattern the Data Insights function could reuse for future use cases.

Big 4 Australian Bank

JUL 2024 – AUG 2024

Gen AI Engineer — Policy Simplification POC

The bank needed to summarise long, dense policy documents without losing their original intent or meaning — a task where getting the fidelity/compression trade-off wrong is costly.

- Deployed LLM API endpoints in Azure ML Studio to summarise lengthy policy documents
- Built a chunking pipeline to break large PDF documents into inputs the endpoint could consume
- Designed evaluation metrics and acceptance criteria to benchmark GPT-4o, GPT-4, GPT-3.5, and Llama 3 on summarisation quality and length reduction

Azure ML Studio

GPT-4o / GPT-4 / GPT-3.5

Llama 3

Python

Impact — Delivered a working POC alongside a model-agnostic evaluation framework the bank could reuse to benchmark future LLMs on document-compression tasks.

Major Australian Property Development Company

JAN 2023 – FEB 2023

Data Strategy Consultant — ML & MLOps Advisory

The client's Data Science team had built predictive ML models but needed help improving, communicating, and deriving ongoing value from them.

- Evaluated the client's existing predictive ML modelling and MLOps processes
- Outlined recommendations to streamline modelling and MLOps workflows in Python
- Delivered a roadmap for more effective stakeholder communication and migration away from legacy systems

Python

MLOps Advisory

Predictive Modelling

Impact — Delivered a technical report and stakeholder presentation that gave the client's data science function a concrete roadmap for maturing its ML practice.

Separate from client delivery — how Bill builds with AI on his own time, including using Claude Code as a daily agentic development partner.

Agentic Development with Claude Code

ONGOING

Bill uses Claude Code as a daily agentic pair-programmer across every personal project he ships — not as an autocomplete tool, but as an agent that plans, edits, runs builds/tests, and drives git/CI end-to-end. This site is itself a live example: it was designed, built, content-audited, and deployed through extended agentic sessions with Claude Code — multi-step planning, tool-orchestrated edits across a whole codebase, automated verification (lint, build, Playwright-driven visual/responsiveness audits), and GitHub Actions deploys, all driven conversationally.

- Comfortable directing multi-step agentic workflows: planning, execution, and verification loops rather than single-shot prompts
- Hands-on with tool-use/orchestration patterns — file edits, shell commands, browser automation, CI/CD — chained together to ship real, deployed product
- Applies the same agentic-coding fluency to iterate quickly across a portfolio of live side projects

Claude Code

Agentic workflows

CI/CD

Playwright

LessonLoop — AI Teaching Assistant

PERSONAL PROJECT

An AI-powered teaching assistant for K-12 educators, built on Claude Opus 4.8 with prompt caching. It keeps persistent memory of each class and generates complete, differentiated daily lesson bundles — plans, slides, worksheets, and exit tickets — so yesterday's results inform tomorrow's instruction.

- Persistent per-class memory via pgvector embeddings — a RAG-style architecture rather than stateless prompting
- Prompt caching with Claude Opus 4.8 to keep latency and cost manageable across large, repeated context
- Structured multi-artefact generation (plans, slides, worksheets, exit tickets) from a single LLM pipeline

Next.js

TypeScript

Claude / LLMs

Drizzle ORM

pgvector



Want to talk through any of these in more detail?

Get in touch — Bill.jyang.r@gmail.com